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From the Combinatorial

model to Constructible Sheaves

Last time The Six-functor formalism on locally compact Haus. spaces, Convolution products

Today Constructing the comparison functor

$$\Psi_\sigma: (\mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp}), \otimes_{\mathrm{Day}}) \longrightarrow (\mathrm{Sh}(M_{\mathbb{R}}), *).$$

Recall if $M \in \mathrm{CMon}(\mathrm{LCH})$ with addition $a: M \times M \rightarrow M$ and zero $0: * \rightarrow M$, then $\mathrm{Sh}(M)$ has a presentably symmetric monoidal structure $*$ called the convolution product with

$$F * G \simeq a_! (pr_1^*(F) \otimes pr_2^*(G)).$$

and unit the skyscraper sheaf $O_!(S) = O_*(S)$. Moreover, the dualizing sheaf $\omega_M \in \mathrm{Sh}(M)$ is naturally a commutative algebra for convolution.

The convolution product of a real vector space

Notation. If X is a topological space, $j: U \hookrightarrow X$ is an open subspace, and $F \in \text{Sh}(U)$, we often write $F \in \text{Sh}(X)$ for $j_!(F)$. In particular, we'll write $\mathbb{1}_U \in \text{Sh}(X)$ for $j_!(\mathbb{1}_U)$.

Similarly, if $i: Z \hookrightarrow X$ is closed and $F \in \text{Sh}(Z)$, we write $F \in \text{Sh}(X)$ for $i_*(F)$.

Prop. Let V be a finite dimensional real vector space and $X, Y \subset V$ polyhedral open subsets. Then there is a natural equivalence

$$\mathbb{1}_X * \mathbb{1}_Y \simeq \mathbb{1}_{X+Y}[-\dim(V)].$$

Here, $X+Y$ denotes the Minkowski sum.

> The proof needs the following general result.

Prop (recognition criterion for locally constant sheaves) Let X be a locally weakly contractible topological space. Then a hypersheaf $L \in \text{Sh}^{\text{hyp}}(X)$ is locally constant if and only if for every inclusion of opens $U \hookrightarrow V$ that is a weak homotopy equivalence, the restriction map

$$L(V) \longrightarrow L(U)$$

is an equivalence

If X is also weakly contractible, then every locally constant hypersheaf is constant.

Proof of convolution formula

Let $a: V \times V \rightarrow V$ denote addition, and note that $X+Y$ is the image of

$$X \times Y \hookrightarrow V \times V \xrightarrow{a} V.$$

Let $j: X \hookrightarrow V$ and $j': Y \hookrightarrow V$ be the inclusions. By definition

$$\mathbb{1}_X * \mathbb{1}_Y = a_! (pr_1^*(j_!(\mathbb{1}_X)) \otimes pr_2^*(j'_!(\mathbb{1}_Y)))$$

By open basechange, we see that

$$\begin{aligned} \mathrm{pr}_1^*(j_!(\mathbb{1}_x)) &\simeq (j \times \mathrm{id})_!(\mathrm{pr}_1^*(\mathbb{1}_x)) \\ &\simeq (j \times \mathrm{id})_!(\mathbb{1}_{x \times v}) \end{aligned}$$

Similarly,

$$\mathrm{pr}_2^*(j'_!(\mathbb{1}_y)) \simeq (\mathrm{id} \times j')_!(\mathbb{1}_{v \times y})$$

We deduce that

$$\begin{aligned} \mathrm{pr}_1^*(j_!(\mathbb{1}_x)) \otimes \mathrm{pr}_2^*(j'_!(\mathbb{1}_y)) &\simeq (j \times j')_!(\mathbb{1}_{x \times y}) \\ &\simeq \mathbb{1}_{x \times y} \end{aligned}$$

Hence

$$\mathbb{1}_x * \mathbb{1}_y \simeq (a|_{x \times y})_!(\mathbb{1}_{x \times y}).$$

It is not hard to show that $a|_{X \times Y} : X \times Y \longrightarrow X + Y$ is a smooth V -bundle. By Poincaré duality, the $!$ -pushforward along addition is locally constant. Since $X + Y$ is contractible, by the prev prop it is constant. Thus the claim reduces to the fact that for any $Z \in \mathcal{L}H$, for the projection $p : Z \times V \longrightarrow Z$ we have

$$p_!(\mathbb{1}) \simeq \mathbb{1}[-\dim(V)]$$

□

Lax Symmetric monoidality of homology

Observation. Let $M \in \mathbf{CMon}(\mathbf{LCH})$. Then the slice category $\mathbf{LCH}/_M$ admits a natural symmetric monoidal structure $\otimes$ with

$$(X, f) \otimes (Y, g) = (X \times Y, f+g)$$

where $f+g: X \times Y \rightarrow M$ is the composite

$$X \times Y \xrightarrow{f \times g} M \times M \xrightarrow{\text{add}} M$$

> The following is a result that is quite technical to prove, but intuitively clear

Thm. Let $M \in \mathbf{CMon}(\mathbf{LCH})$. There is a lax symmetric monoidal functor

$$\Gamma_M: (\mathbf{LCH}/_M, \otimes) \longrightarrow (\mathbf{Sh}(M), *)$$

given on objects by

$$(X, f) \longmapsto f_! f^! (\omega_M)$$

homology of X
over M with
coeffs in ω_M

The assignment on morphisms sends

$$\begin{array}{ccc} X & \xrightarrow{p} & Y \\ & \searrow f & \swarrow g \\ & M & \end{array}$$

to

$$f_! f^! (\omega_M) \simeq g_! (p_! p^!) g^! (\omega_M) \xrightarrow{\text{counit}} g_! g^! (\omega_M)$$

The homology of a cone

Recall For N a lattice, $\text{Poly}(M)$ is the poset of subsets of $M_{\mathbb{R}}$ of the form $(\text{cone}) + (\text{polytope})$. For $\sigma \in N_{\mathbb{R}}$ a cone, we write

$$\begin{aligned}\Theta(\sigma) &= \{m + \sigma^{\vee} \mid m \in M\} \\ &\subset \text{Mod}_{\sigma^{\vee}} \text{Poly}(M)\end{aligned}$$

> we'll start by analyzing

$$\Gamma_{M_{\mathbb{R}}} \cdot \text{Poly}(M) \longrightarrow \text{Sh}(M_{\mathbb{R}}).$$

Prop. Let N be a lattice and $\sigma \in N_{\mathbb{R}}$ a cone. Then the sheaf

$$\omega_{\sigma^{\vee}} \simeq \Gamma_{M_{\mathbb{R}}}(\sigma^{\vee})$$

on $M_{\mathbb{R}}$ is an idempotent algebra with respect to convolution,

> First we state a corollary.

Cor. Let N be a lattice. Then the assignment $\sigma \mapsto \omega_\sigma$ defines a functor

$$\text{Cone}(N_{\mathbb{R}})^{\text{op}} \longrightarrow \text{CAlg}^{\text{idem}}(\text{Sh}(M_{\mathbb{R}}), *)$$

Proof:

Recall that for a symmetric monoidal ∞ -category $\mathcal{C}$, the full subcategory $\text{CAlg}^{\text{idem}}(\mathcal{C}) \subset \text{CAlg}(\mathcal{C})$ is a poset, and there is a map of idem algebras $A \rightarrow B$ if and only if B is an A -algebra. So the functoriality can be specified at the 1-categorical level, which is obvious. $\square$

> For the proof of the prop we need:

Lemma. Let N be a lattice and $\bar{U} \subset M_{\mathbb{R}}$ be a closed polyhedral subset of top dimension with interior U . Then the map of homology sheaves

$$\Gamma_{M_{\mathbb{R}}}(\mathcal{U}) \longrightarrow \Gamma_{M_{\mathbb{R}}}(\bar{\mathcal{U}})$$

is an equivalence.

> The proof uses the following:

Proof of Lemma

Since every sheaf on $M_{\mathbb{R}}$ is a hypersheaf, it suffices to check that this map induces an equivalence on stalks.

If $x \notin \partial \bar{U}$, this is clear by proper base change: both agree with the restriction to U on U and are 0 outside of $\bar{U}$.

For $x \in \partial \bar{U}$, the stalk of the left-hand side vanishes. So we just need to check this for the right-hand side. To compute the stalk, let $(D_k)_{k \geq 0}$ be disks of shrinking radii centered at x . Then

$$\Gamma_{M_{\mathbb{R}}}(\bar{U})_x \simeq \operatorname{Colim}_{k \rightarrow \infty} \Gamma_{M_{\mathbb{R}}}(\bar{U})(D_k)$$

Using that $\omega_{M_{\mathbb{R}}} \simeq \mathbb{I}[n]$ for $n = \dim(M_{\mathbb{R}})$ and applying proper basechange we see that for $i: \bar{U} \hookrightarrow M_{\mathbb{R}}$ the inclusion, we have

$$\Gamma_{M_R}(\bar{U})(D_k) \simeq (i_! i^*(\mathbb{1}[\hbar]))(D_k)$$

$$\simeq \text{fib}(\mathbb{1}(D_k) \longrightarrow \mathbb{1}(D_k - \bar{U}))[\hbar].$$

Since $\bar{U}$ is polyhedral, for a sufficiently small ball D , $D \setminus \bar{U}$ is also contractible. Hence for $k \gg 0$, by local constancy the map is an equivalence, so the fiber is zero. $\square$

Combinatorial to constructible comparison

Observe. If (N, Σ) is a fan, then the assignment

$$\sigma \mapsto \text{Mod}_{\omega_{\sigma\nu}} \text{Sh}(M_{\mathbb{R}})$$

with functoriality given by tensoring defines a diagram

$$\Sigma^{\text{op}} \longrightarrow \text{CAlg}(\text{Pr}^{\text{L}})$$

Taking limits, we obtain a symmetric monoidal left adjoint

$$L_{\Sigma}: \text{Sh}(M_{\mathbb{R}}) \longrightarrow \lim_{\sigma \in \Sigma^{\text{op}}} \text{Mod}_{\omega_{\sigma\nu}} \text{Sh}(M_{\mathbb{R}})$$

Write R_{Σ} for the lax symmetric monoidal right adjoint to L_{Σ} , induced by the fully faithful forgetful functors

$$\text{Mod}_{\omega_{\sigma\nu}} \text{Sh}(M_{\mathbb{R}}) \longrightarrow \text{Sh}(M_{\mathbb{R}})$$

Since each of these functors is fully faithful, R is fully faithful.

Construction Let N be a lattice and $\sigma \subset N_{\mathbb{R}}$ a cone. Write Ψ_{σ} for the composite

$$\Theta(\sigma) \longrightarrow \text{Mod}_{\sigma^{\vee}} \text{Poly}(M) \xrightarrow{\Gamma_{M_{\mathbb{R}}}} \text{Mod}_{\omega_{\sigma^{\vee}}} \text{Sh}(M_{\mathbb{R}})$$

Recall that the first functor is symmetric monoidal, and the second is lax symmetric monoidal.

Check. The functor $\Psi_{\sigma} : \Theta(\sigma) \longrightarrow \text{Mod}_{\omega_{\sigma^{\vee}}} \text{Sh}(M_{\mathbb{R}})$ is symmetric monoidal.

> Let $n = \dim(M_{\mathbb{R}})$.

> This amounts to our earlier result that for polyhedral subsets $X, Y \subset M_{\mathbb{R}}$, we have

$$\mathbb{1}_X * \mathbb{1}_Y \simeq \mathbb{1}_{X+Y}[-n]$$

> The point is that for $X \in \Theta(\sigma)$ an integral translate of a cone, X is homeomorphic to a Euclidean space, so by Poincaré duality, $\omega_X \simeq \mathbb{1}_X[n]$

> Hence for $X, Y \in \Theta(\sigma)$ we have

$$\omega_X * \omega_Y \simeq (\mathbb{1}_X[n]) * (\mathbb{1}_Y[n])$$

$$\simeq (\mathbb{1}_X * \mathbb{1}_Y)[2n]$$

$$\simeq (\mathbb{1}_{X+Y}[-n])[2n]$$

$$\simeq \mathbb{1}_{X+Y}[n]$$

$$\simeq \omega_{X+Y}.$$

Using the universal property of Day convolution, we write

$$\Psi_\sigma: (\text{Fun}(\Theta(\sigma), \text{Sp}), \otimes_{\text{Day}}) \longrightarrow (\text{Mod}_{\omega_\sigma} \text{Sh}(\mathcal{M}_R), *)$$

for the symmetric monoidal left adjoint induced by Ψ_σ .

Check (slightly technical). Given a fan (N, Σ) , the functors

$(\Psi_\sigma)_{\sigma \in \Sigma^\circ}$ assemble into a natural transformation.

$$\mathrm{Fun}(\Theta(-)^{\circ p}, S_p) \longrightarrow \mathrm{Mod}_{\omega_{(-)^{\vee}}} \mathrm{Sh}(M_R)$$

of diagrams $\Sigma^{\circ p} \longrightarrow \mathrm{CAlg}(R^L)$.

Notation. For a fan (N, Σ) , write

$$\Psi_{\Sigma} \lim_{\sigma \in \Sigma^{\circ p}} \mathrm{Fun}(\Theta(\sigma)^{\circ p}, S_p) \longrightarrow \lim_{\sigma \in \Sigma^{\circ p}} \mathrm{Mod}_{\omega_{\sigma^{\vee}}} \mathrm{Sh}(M_R)$$

for the induced symmetric monoidal functor

Def Let (N, Σ) be a fan. The coherent-constructible correspondence

$$K_{\Sigma}(\mathrm{QCoh}(X_{\Sigma}/T), \otimes) \longrightarrow (\mathrm{Sh}(M_R), *)$$

is the lax symmetric monoidal functor given by the composite

$$\mathrm{QCoh}(X_{\Sigma}/T) \xrightarrow[\sim]{\Phi_{\Sigma}^{-1}} \lim_{\sigma \in \Sigma^{\circ p}} \mathrm{Fun}(\Theta(\sigma)^{\circ p}, S_p) \longrightarrow \lim_{\sigma \in \Sigma^{\circ p}} \mathrm{Mod}_{\omega_{\sigma^{\vee}}} \mathrm{Sh}(M_R) \xrightarrow{\int R_{\Sigma}} \mathrm{Sh}(M_R)$$

Note that the only functor here which we have not seen is symmetric monoidal is R_Z

Next time When Z is smooth projective, R_Z is an equivalence, so k_Z is symmetric monoidal